

Capstone Project Proposal

FA 550: Data Visualization

Real Estate Market Efficiency & Pricing Insights Dashboard

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Project Overview

Track Selection

This project will follow the **Executive Dashboard** track. The goal is to build an interactive dashboard that analyzes real estate transactions and supports data-driven decision-making.

Project Title and Description

Real Estate Market Efficiency & Pricing Insights Dashboard

This project will develop a multi-page dashboard that analyzes real estate property transactions. The focus is on comparing sale prices with estimated property values to identify pricing patterns, trends, and potential inefficiencies in the market. The dashboard will include multiple interconnected visualizations that allow users to explore data over time, across property types, and by location (if available).

Motivation

Real estate decisions are often made with limited or fragmented information. This project aims to provide a clear, data-driven view of how properties are priced relative to their estimated values. Understanding these patterns can help identify market inefficiencies and support better decision-making for buyers and investors.

Goals and Objectives

The main goal is to design a professional and user-friendly dashboard that presents meaningful insights about real estate transactions. The objectives include:

- Analyze the relationship between sale price and estimated value
- Identify pricing trends over time
- Detect overvalued and undervalued properties
- Compare performance across property types
- Present insights through clear and interactive visualizations

Target Audience

The dashboard is intended for real estate investors, home buyers, and analysts who want to better understand housing market trends and pricing behavior.

Scope

This project focuses on analyzing real estate property transactions to understand pricing behavior and market efficiency. The analysis will include comparisons between sale prices and estimated values, trends over time, and differences across property types and locations.

The scope is limited to the available dataset and its variables, including property features such as number of rooms, bathrooms, and area. Advanced predictive modeling or external economic factors will not be included in this phase, as the focus is on exploratory data analysis and interactive visualization.

The final deliverable will be a multi-page interactive dashboard that provides clear, actionable insights and supports user-driven exploration.

Risk Analysis

Several potential risks may impact the successful completion of this project. These include data quality issues such as missing or inconsistent values, which may affect the accuracy of analysis. Additionally, limited geographic detail or incomplete records could restrict certain types of insights.

There is also a risk related to dashboard performance when working with a large dataset, which may affect responsiveness. To mitigate these risks, data will be carefully cleaned and validated, unnecessary fields will be removed, and visualizations will be optimized for performance. The project scope will remain flexible to ensure timely completion if any constraints arise.

Data Plan

Data Sources

The primary dataset for this project will be the **Real Estate Property Transactions Dataset** obtained from Kaggle. This dataset contains detailed records of property transactions, including sale prices, estimated values, property types, and transaction dates.

Data Description

The dataset contains approximately 10,000 property transaction records, providing sufficient data for meaningful analysis across time, location, and property characteristics. The dataset includes key variables required for analysis:

- Sale price of each property
- Estimated or predicted property value
- Property type and related attributes
- Transaction date
- Geographic information including locality, enabling location-based analysis

The dataset is expected to contain a sufficient number of records to support trend analysis and comparisons across different categories.

Access Verification

The dataset has been accessed and reviewed through Kaggle. Initial inspection confirms that it contains the necessary variables, particularly sale price and estimated value, which are essential for analyzing pricing efficiency.

Anticipated Challenges

- Missing or incomplete values in key fields
- Inconsistent formatting of dates or categories
- Presence of outliers that may affect analysis
- Limited geographic detail for location-based insights

Data Preparation Plan

To prepare the dataset for analysis, the following steps will be performed:

- Clean missing or invalid data
- Utilize property features such as number of rooms, bathrooms, and carpet area for deeper analysis
- Standardize formats for dates and categorical variables
- Identify and handle outliers
- Create new variables, including:
 - Price difference (Sale Price - Estimated Value)
 - Percentage difference to measure overvaluation or undervaluation
- Aggregate data where needed for time-based and category-based analysis

Technical Plan

Tools and Technologies

The dashboard will primarily be developed using **Power BI** due to its strong capabilities in building interactive, multi-page dashboards and handling large datasets efficiently. Power BI also provides built-in tools for data transformation and KPI-based analysis, making it well-suited for this project.

As an alternative, **Tableau** may be considered for visualization design if needed, particularly for enhancing visual presentation. However, Power BI will be the main tool used for development.

Visualization Plan

The dashboard will include multiple visualizations to effectively present real estate insights:

- KPI cards such as average sale price, average estimated value, and price difference
- Comparative charts analyzing price based on property features such as rooms, bathrooms, and area
- Line charts to show pricing trends over time
- Scatter plot to compare sale price vs. estimated value
- Bar charts to compare property types and transaction counts
- Distribution charts to analyze pricing differences
- Map visualization (if location data is available)

These visualizations will be organized across multiple dashboard pages to maintain clarity and usability.

Visualizations will be optimized to ensure smooth performance with large datasets.

Interaction Design

The dashboard will include interactive features to enhance user experience:

- Filters for date range, property type, and location
- Drill-down capabilities for exploring data at different levels (e.g., region to city)
- Hover tooltips to provide additional context for data points

These interactions will allow users to explore the dataset dynamically and gain deeper insights.

Wireframes / Mockups

The dashboard will be structured into multiple pages:

- Overview page with key metrics and general trends
- Pricing analysis page focusing on sale vs. estimated value
- Trends page showing changes over time
- Comparison page analyzing property types or regions

A simple layout will be designed before development to ensure clarity and logical flow.

AI Strategy

AI tools such as ChatGPT/Claude will be used throughout the project to assist with:

- Generating ideas for visualizations and dashboard structure
- Debugging issues during data preparation and visualization
- Improving design and layout decisions
- Assisting with documentation and explanations

All AI-generated outputs will be reviewed and validated before use.

Technical Risks

Potential risks include:

- Data quality issues such as missing or inconsistent values
- Limited location data affecting geographic analysis
- Performance issues with large datasets

To address these risks, data will be cleaned carefully, unnecessary fields will be removed, and visualizations will be optimized for performance.

Timeline & Milestones

The project will be completed over a four-week period with the following milestones:

Week 1: Data Acquisition and Exploration

- Finalize dataset selection and confirm access
- Perform initial data exploration and understanding
- Identify key variables and potential insights

Week 2: Data Preparation and Initial Visualizations

- Clean and preprocess the dataset (handle missing values, format data)
- Create derived variables such as price difference and percentage difference
- Develop initial visualizations and dashboard structure

Week 3: Dashboard Development and Refinement

- Build full dashboard with multiple pages and visualizations
- Add interactivity including filters and drill-down features
- Improve layout, design, and usability

Week 4: Testing and Finalization

- Test dashboard functionality and performance
- Make final adjustments and improvements
- Complete documentation and prepare final submission

Success Criteria

The success of this project will be evaluated based on the following measurable criteria:

- The dashboard includes at least 8 interconnected visualizations across multiple pages
- Key insights about pricing trends and market efficiency are clearly communicated
- Users can easily navigate and interact with the dashboard using filters and drill-down features
- The dashboard performs efficiently and loads within a reasonable time
- The dashboard reveals at least 3 meaningful and actionable insights about pricing trends or market efficiency
- The final product is visually clear, professional, and suitable for inclusion in a portfolio

AI Disclosure Statement

I pledge on my honor that I have not given or received any unauthorized assistance on this assignment/examination. I further pledge that I have not copied any material from a book, article, the Internet or any other source except where I have expressly cited the source. I have documented all AI tool usage including prompts and tool selection rationale.

Signature : Aayush Gandhi

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